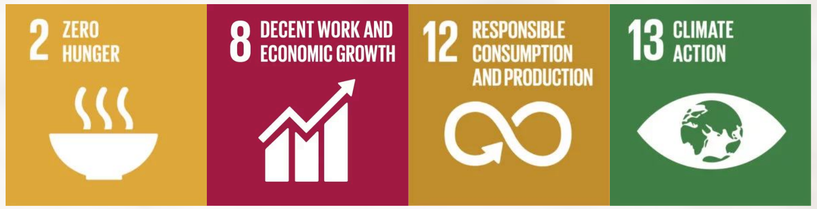
Farmers' Smile v1.0

Know what to plant. Know when to plant.



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The Problem

Farmers still rely on traditional planting calendars, but rainfall and temperature are becoming less predictable. This can lead to wrong crop choices, poor timing, wasted inputs, and lower returns.

Images From Internet

This problem is already happening in Asia.

Reuters

World Business Markets Sustainability Legal Commentary Technology Investig

Hot weather hurts Asian crops as powerful El Nino takes shape

By **Naveen Thukral**

June 4, 2026 12:12 PM GMT+7 · Updated June 4, 2026

Food and Agriculture Organization of the United Nations

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FAO Regional Office for Asia and the Pacific

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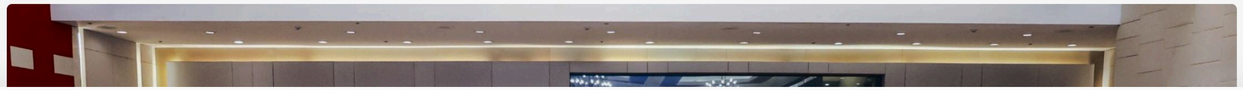
Southeast Asian countries turn ambition into action with USD 2.6 billion in proposed climate-smart agriculture investments

As climate change reshapes agrifood systems across the region, countries are working to turn ambition into investment and planning into action.



Thai PBS WORLD

HOME NEWS PROGRAMMES GALLERY INFOGRAPHIC QUOT

SHARE

Thailand's leading environmental expert Sonthi Kochawat has warned that the country is entering a Super El Niño phase, which will bring extremely hot weather and a prolonged dry season, leading to drought that could severely impact the agricultural sector.

He noted that outdoor temperatures yesterday and today averaged 38°C, and may reach 39°C. With relative humidity exceeding 60%, the heat index could rise to about 58.7°C—an extremely dangerous level that can lead to heatstroke.

Sonthi explained that the extreme heat may cause the body's temperature to rise to 40°C within just 10–15 minutes of exposure to sunlight.

Thai PBS WORLD

HOME NEWS PROGRAMMES GALLERY INFOGRAPHIC QUOT

SHARE

Many rice farmers in Thailand's central province of Chainat are caught in a dilemma over how to proceed with the cultivation of their main rice crops against a background of the unpredictable weather conditions.

On the one hand is the prospect of the premature arrival of drought, forecast to arrive in September and last through October. On the other is the warning from the Royal Irrigation Department of the arrival of a mass of water from the northern region into the Chao Phraya River, which will force an increase in the discharge of water through the Chao Phraya Dam.

The Farmer's Decision

"What should I plant?"

"When should I plant?"

"Will this crop survive this season?"

Images From Internet

Our Solution

Existing approach:

"When should I plant this crop?"

Farmers' Smile:

"Given the expected climate conditions, what should I plant and when?"

Our Prototype

How It Works in the Backend

Input

- 30+ Years Historical Climate Data
- Current Weather Data
- Seasonal Forecast Data
- Crop Requirements Data
- Local Farm Conditions Data

Process

Small AI

Output

- What should I plant?
- When should I plant?
- What are the risks?
- Why does the AI recommend it?

Deliver

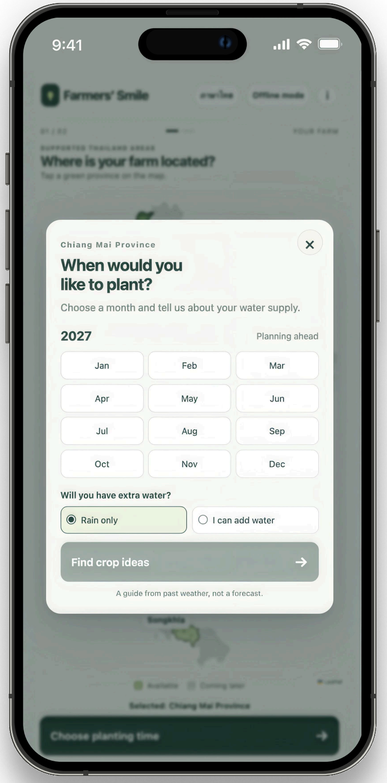
- Lightweight app
- SMS
- Local-language voice

Our Prototype

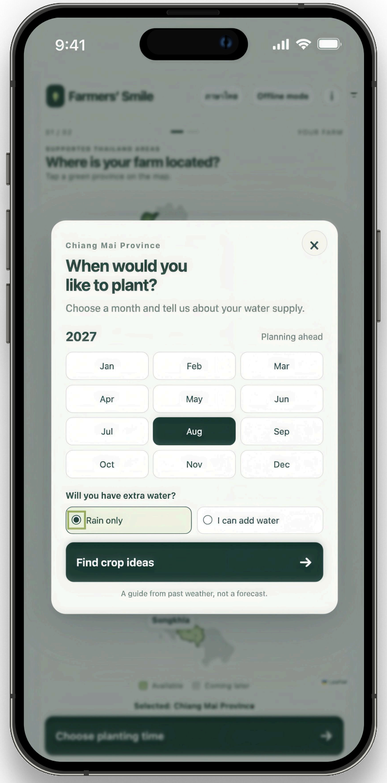
Product Demo

A farmer compares crops in one flow.

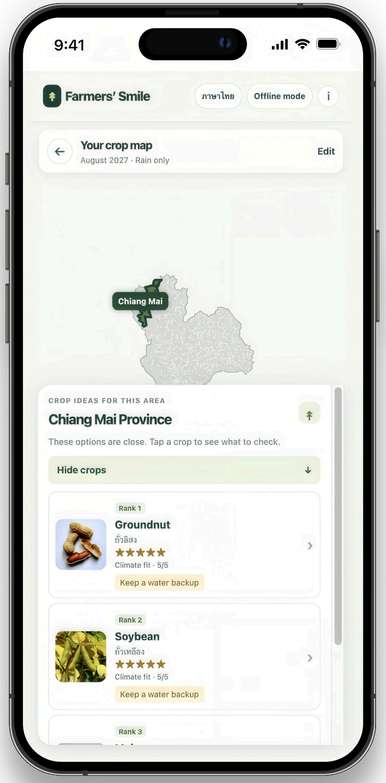
Select farm location



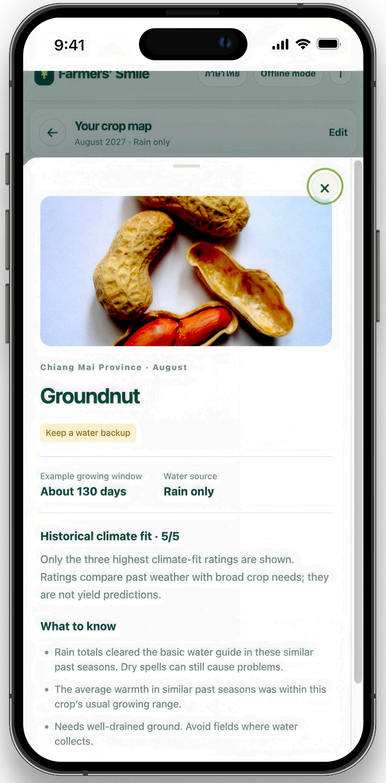
Choose planting month



Enter water condition



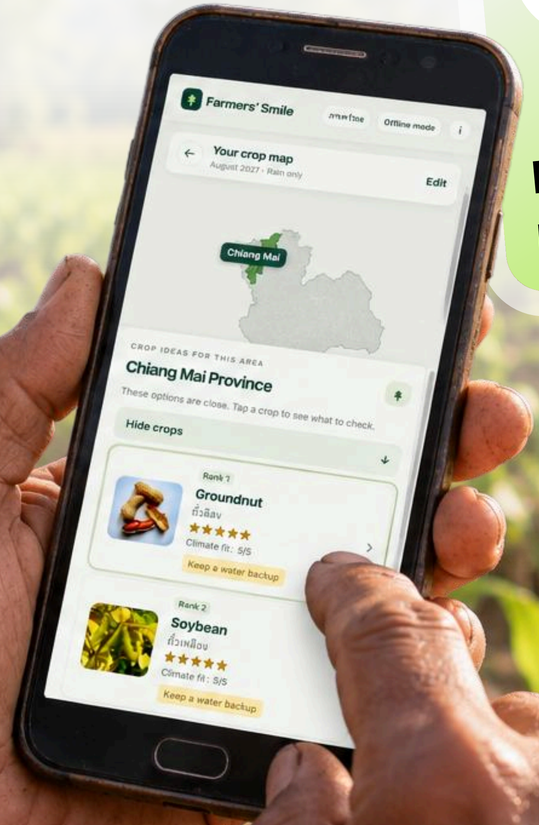
Compare 3 crops



Read the recommendation

Why Small AI?

Designed for farmers with limited connectivity



Lightweight model

Offline-capable

Low Cost Development

Local crop data

Local-language delivery.

Impact we plan to test.

Better decisions can reduce avoidable farming risk.

Reduction in
Climate-Inappropriate Crop Selection

Reduction in
Planting-Window Mismatch

Reduction in
Avoidable Input Expenditure

Increase in
Climate-Suitable Crop Selection

Team

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THANK YOU

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University:

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Hat Yai Campus

Useful With or Without Internet

ONLINE

Latest climate data
Fetch updated weather and seasonal forecasts.

Full AI analysis
Run updated crop suitability analysis.

Data synchronization
Download the newest crop and climate information.

Detailed maps
Access richer location and environmental information.

Model updates
Receive updated recommendation models.

OFFLINE

Saved recommendations
View previously generated crop recommendations.

Local crop database
Access essential crop data on device.

Basic analysis
Run lightweight calculations locally.

Farm records
Save location, crop choices, and recommendations.

Sync later
Automatically update when internet access returns.

Future Development Plan

Phase 1 — Prototype

- Chiang Mai, Songkhla, Surat Thani
- 1996–2025 weather history
- Thai/English app
- Offline demo
- Start outreach

Phase 2 — Local Adoption

- Run partner-supported farmer demos
- Help first-time users get started
- Gather farmer and adviser feedback
- Improve local data and guidance
- Validate before wider use

Phase 3 — Regional Growth

- Add areas after local validation
- Develop locally reviewed crop guidance
- Expand language and partner support
- Explore other Asian regions

NOW

NEXT

FUTURE

Business Model Canvas

KEY PARTNERS

- Local government (agriculture & research departments)
- Agricultural universities & research institutions
- Meteorological & Climate data providers
- AI/ technology partners
- Farmer cooperatives
- NGO & CSR partners

KEY ACTIVITIES

- Collect, clean and process
- climate & agricultural data
- Develop & improve AI model
- Generate recommendations
- Prototype testing & field validation
- App maintenance & user support
- Farmer onboarding & outreach

VALUE PROPOSITIONS

For Farmers (Know what to plant, Know when to plant)

- Climate-informed crop recommendations
- Localized planting guidance
- Simple & accessible mobile app
- Reduced risk from climate uncertainty

For Organizations (A salable platform for delivering climate-smart agricultural support.)

- Support many farmers through one platform
- Regional climate & agriculture insights
- Data analytics & reporting

CUSTOMER RELATIONSHIPS

- Self-service mobile app
- Automated recommendation & alerts
- Customer support
- Long-term partnership contracts
- Co-creation with local organizations

CUSTOMER SEGMENTS

Users (Free Service)

- Smallholder farmers

Paying Customers

- Farmer cooperatives
- Agricultural organizations
- Local government
- NGOs / development organizations

Partners / Expert Users

- Agricultural professionals
- Researchers & agronomists

KEY RESOURCES

- Climate & weather data
- Crop & agricultural research data
- AI recommendation model
- Skilled technical team
- Agricultural expertise
- APP / platform infrastructure
- Farmer / user data (with consent)

COST STRUCTURE

- App development & maintenance
- Cloud & hosting costs
- Data & API costs
- AI / R&D
- Marketing & farmer outreach
- Employee salaries
- Validation & field testing

CHANNELS

- Mobile app (iOS/Android)
- SMS/messaging alerts
- Farmer cooperatives
- Government / NGO programs
- Social media (TikTok, Facebook, IG)
- Radio/local media
- Newspaper & community events

REVENUE STREAMS

- B2B/B2G Subscriptions
- Organizational Projects
- Optional Farmer Premium
- Future API / Data Services

Designed for Low Connectivity and Limited Digital Experiences

- Step-by-step user guide
- Clear, simple flow
- Thai or English
- Works offline after download

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